

$$\Omega_0 = 2^{2N_A} \quad P_O = \frac{1}{2^4} \binom{2}{4} = \frac{3}{8}$$

$$\Omega = \Omega_0 \cdot (P_O)^{N_A} = \left(\frac{3}{2}\right)^{N_A}$$

$$S = k_B \ln \Omega = \boxed{\frac{3}{2} R \approx 3.37 \text{ J mol}^{-1} \text{ K}^{-1}}$$